

# Control Barrier Functions

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## ABSTRACT

Exploring the Control Barrier Functions using gaussian kernels. Mainly debugging why they did not produce the expected outputs. Once the error was found, testing different length scales in order to determine the optimal size for running on the indoor track.

Keywords: Keyword1, Keyword2, Keyword3

## INTRODUCTION

Control Barrier Functions(CBFs) are functions that can guarantee the safety of the vehicle using lyapunov analysis. The CBFs used are based on the paper ?. This paper demonstrates the use of Gaussian kernels to construct barrier functions in real time. The reason for the use of Gaussian kernels is to create smooth differentiable barriers. The barriers are defined for all locations making feasibility of the barriers more probable.

## METHODS AND MATERIALS

Guidelines can be included for standard research article sections, such as this one.

## SOME L<sup>A</sup>T<sub>E</sub>X EXAMPLES

Use section and subsection commands to organize your document. L<sup>A</sup>T<sub>E</sub>X handles all the formatting and numbering automatically. Use ref and label commands for cross-references.

### Figures and Tables

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| Item         | Quantity |
|--------------|----------|
| Candles      | 4        |
| Fork handles | ?        |

**Table 1.** An example table.

### Mathematics

L<sup>A</sup>T<sub>E</sub>X is great at typesetting mathematics. Let  $X_1, X_2, \dots, X_n$  be a sequence of independent and identically distributed random variables with  $E[X_i] = \mu$  and  $\text{Var}[X_i] = \sigma^2 < \infty$ , and let

$$S_n = \frac{X_1 + X_2 + \dots + X_n}{n} = \frac{1}{n} \sum_i^n X_i$$

denote their mean. Then as  $n$  approaches infinity, the random variables  $\sqrt{n}(S_n - \mu)$  converge in distribution to a normal  $\mathcal{N}(0, \sigma^2)$ .



**Figure 1.** An example image of a frog.

### **Lists**

You can make lists with automatic numbering ...

1. Like this,
2. and like this.

... or bullet points ...

- Like this,
- and like this.

... or with words and descriptions ...

**Word** Definition

**Concept** Explanation

**Idea** Text

## **ACKNOWLEDGMENTS**

Additional information can be given in the template, such as to not include funder information in the acknowledgments section.